Sumeet Parameshwar Shirgure

PhD Student · Quantum Computing

University of Central Florida

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Education

University of Central Florida — *Orlando, FL*

PhD in Computer Engineering — *August 2025 - Present*

- Advisor: Dr. Siyuan Niu
- Research focus : Quantum Computing and Quantum Information, Computer Architecture

University of Southern California — *Los Angeles, CA*

MS in Computer Science — *September 2021 - May 2023*

- Supervisor: Dr. Massoud Pedram
- Project description : VLSI EDA tools for Superconductor Electronic Logic

Indian Institute of Technology — *Kharagpur, WB*

B.Tech(Hons.) + M.Tech in Computer Science and Engineering — *July 2015 - April 2020*

- Graduated with honors, and a cumulative GPA of 9.06/10.00

Professional Experience

Ming Hsieh Department of Electrical and Computer Engineering, USC — *Los Angeles, CA*

Research assistant — *May 2022 - April 2023*

- Development of EDA tools for superconductor electronics

Google — *Bengaluru, India*

Software Engineer — *August 2020 - June 2021*

- Software Engineer in the call ads infrastructure team

D.E Shaw — *Hyderabad, India*

Software Engineering intern — *May 2019 - July 2019*

- Built machine learning infrastructure

Adobe — *Noida, India*

Deep learning research intern — *May 2018 - July 2018*

- Research on neural language models using recurrent neural networks

Awards, Fellowships, & Grants

2015	**Ranked 415th among 1.5 million applicants** , IIT Joint Entrance Exam (Advanced)	*India*
2019	**Represented India at World Finals**, International Collegiate Programming Contest	*Porto*
2019	**Top 1000 participants worldwide**, Google's CodeJam	*Online*
2019	**Top 500 participants worldwide**, Meta's HackerCup	*Online*
2021	**Dean's Master's scholarship**, $5000 fee waiver	*USC*
2022	**Quantum computing hackathon winner** , QCHack	*Online*
2022	**Accepted into the CS MS honors program**,	*USC*
2022	**Graduate RA award**, Full tuition fee waiver and stipend	*USC*
2023	**Initiated into Phi Kappa Phi**, Top 10% of graduate students	*USC*
2023	**MIT iQuHACK participant**, Invited to the in-person quantum computing hackathon	*MIT*
2025	**ORCGS Doctoral Fellowship**, Full tuition fee waiver and stipend	*UCF*

Publications

Published

Sumeet Shirgure and Siyuan Niu
Error Mitigation in Dynamic Circuits for Hamiltonian Simulation
In Proceedings of the Great Lakes Symposium on VLSI 2026 (GLSVLSI '26). Association for Computing Machinery, New York, NY, USA, 843–849

Sumeet Shirgure, Efekan Kökcü, Anupam Mitra, Wibe Albert de Jong, Costin Iancu, Siyuan Niu
Scalable Benchmarking Framework for Dynamic Quantum Circuits.
To appear in 59th IEEE/ACM International Symposium on Microarchitecture

In Review / In Preparation

Sumeet Shirgure, Efekan Kökcü, Siyuan Niu
Challenges in Barren Plateau Mitigation with Dynamic Parameterized Quantum Circuits
arXiv preprint arXiv:2606.23751 (2026). With the editors of *Quantum*

Sumeet Shirgure, Yanzhang Zhu, Di Wu, Siyuan Niu
Speculative Matching for Low Latency Surface Code Quantum Error Correction
(In preparation, 2026)

Sumeet Shirgure, Siyuan Niu
Fast Bounds on Matching Decoder Soft Output for the Surface Code
(In preparation, 2026)

Sumeet Shirgure, Siyuan Niu
The Syndrome Buffer Sizing Problem in Quantum Error Correction
(In preparation, 2026)

Presentations

Contributed Talks

Scalable Benchmarking Framework for Dynamic Quantum Circuits
59th IEEE/ACM International Symposium on Microarchitecture, November 2026, Athens, Greece

Challenges in Barren Plateau Mitigation with Dynamic Parameterized Quantum Circuits
Quantum Techniques in Machine Learning, December 2026, Stellenbosch University, South Africa

Poster Presentations

A flaw in the NeurIPS unlearning challenge and an algorithmic framework for entropy regularization
Neural Information Processing Systems, December 2023, New Orleans, LA

Dynamic Circuit Compilation for Sparse Qubit Connectivity
IEEE International Conference on Quantum Computing and Engineering, September 2025, Albuquerque, USA
Adaptive Quantum Circuits, November 2025, Boston MA, USA

Extracurricular Activities

Service and Outreach

2015-2017 **National Service Scheme**, Student Volunteer *IIT Kharagpur*
2026 **Qiskit Fall Fest 2026 at UCF**, Organizer *UCF*

Peer Review

Sub-reviewer for GLS VLSI 2026